AN EMPIRICAL BRIDGE FROM REALIZABLE KARP FACTOR TO M-STEP FOSTER DRIFT ON RESIDUE QUOTIENTS OF THE COLLATZ MAP

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ABSTRACT. We report a finite empirical diagnostic for the accelerated odd Collatz map $S(n) = (3n+1)/2^{v_2(3n+1)}$. The diagnostic chain combines three saved computational audits: a bounded simple-cycle realizability audit on finite LTE-closed residue graphs, a phase-decomposed renewal drift audit, and a Foster-Lyapunov drift audit for $V(n) = \log_2(n) + v_2(n+1)$ on residue quotients. At the tested levels $(q, R_{\max}) = (5,4), (6,5), (7,6)$, the only positive-integer simple-cycle realizer in the bounded audit is the elementary valuation word [2], giving realizable Karp factor 3/4. The finite renewal-window sweep is consistent with the phase identity

$$\mu_{\text{step}} = \frac{1}{2}\log_2(3/2) + \frac{1}{2}\log_2(3/4) = \log_2(3/4),$$

with post-exit valuation deviation shrinking from -0.03473021694546086 to -0.006503489706314536. The same candidate Lyapunov function has residue-uniform negative drift at the tested grid value $m = 16$, with margin $\epsilon = 0.1$, over the sampled windows $[10^2, 10^4]$ through $[10^{12}, 10^{15}]$. No proof claim is made: the artifacts do not control arbitrary infinite switching paths and do not give deterministic per-orbit descent. Combining this framework's Foster condition with Chang's recent structural reduction (`arXiv:2603.25753`) gives a joint argument whose three named quantitative verifications are empirically closed at finite windows, leaving the residual exactly the Tao-style distributional-to-pointwise wall. This is a finite empirical diagnostic; the Collatz conjecture remains open.

1. Introduction

This paper records the public version of a closed audit chain: finite tail graphs, cycle realizability, phase-decomposed renewal drift, and Foster-Lyapunov residue diagnostics. It is in the genre of computational mathematics and experimental mathematics. It makes no claim of proving the Collatz conjecture, no descent theorem, and no theorem about deterministic per-orbit descent.

The object is the accelerated odd Collatz map

$$S(n) = \frac{3n+1}{2^{v_2(3n+1)}},$$

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applied to odd positive integers. The project began as a certificate-search framework and ended as a reproducible finite diagnostic framework. The audit history in `docs/PROJECT_JOURNEY.md` records several demoted claims: a per-step running-debt Lyapunov ansatz failed at roughly 50 percent non-decrease, a structural Diophantine exclusion was a resolution artifact, and an early closed-form reading of the renewal constant was rejected.

The current finite picture is narrower. On tested LTE-closed tail graphs, bounded integer-realizable simple cycles are filtered down to the elementary [2] cycle, whose factor is $3/4$. Independently, the phase-decomposed renewal drift audit finds the same value as a mean-step target after separating tail-internal and post-exit phases. Finally, the Foster audit finds that

$$V(n) = \log_2(n) + v_2(n+1)$$

does not give one-step residue-uniform negativity, but does give an m-step finite residue diagnostic at $m = 16$ on the tested windows.

There are now two relevant Chang 2026 papers. The human-LLM collaboration paper [Cha26b] is a methodological and structural companion, while the later structural reduction [Cha26a] reduces the Collatz question to a fixed-modulus, one-bit orbit-mixing problem. The compatibility analysis in `docs/reports/chang_2603_25753_compatibility.md` reads the two efforts as addressing the same distributional-to-pointwise wall from complementary angles: Chang isolates a one-bit balance bottleneck, and this framework supplies an $m = 16$ Foster residue diagnostic that can be composed with that reduction as a joint argument.

Caveat 1. *All quantitative statements below are tied to saved JSON artifacts. The language "finite empirical diagnostic" is intentional: the computations are finite, sampled or bounded, and do not imply a global statement for all positive integers.*

2. Finite Setup and Artifact Discipline

Odd integers in a tail are represented as

$$n = 2^R u - 1,$$

with state coordinate (R, u) and bounded unit precision. The LTE closure compresses deeper-tail overflow back to the tracked maximum tail depth by following the forced valuation-one run. At the largest saved LTE-closed level reported in the note, $(q, R_{\max}) = (10, 7)$ has 3584 states, 8704 edges, and 44 closed overflow edges `docs/reports/tail_aware_lte_closed_projective_jsr.json`.

The finite graph still contains abstract high-growth switching. The saved worst-case switching value at that level is JSR factor

$$1.2305030340113887,$$

with max cycle mean 0.29924821500679044 in $\log_2$ units `docs/reports/tail_aware_lte_closed_projec`
In contrast, the lift-count Markov model on the same level has one recurrent

component with 1113 states and average $\log_2$ growth

$$-0.4118668182695867,$$

or typical factor 0.7516501240662575 per accelerated step `docs/reports/tail_aware_markov_lyapunov`

This worst-versus-typical split is the reason for the audit discipline. The finite operator data are not read as a proof object. They are paired with cycle-realizability, renewal, and residue-drift diagnostics.

Definition 2. *For an odd n, write $R(n) = v_2(n+1)$. The phase-aware Lyapunov candidate used below is*

$$V(n) = \log_2(n) + R(n).$$

3. Realizable Karp Factor

The bounded-cycle audit separates abstract cycles in the finite tail graph from cycle words that lift to positive integer orbits. In the joint Karp/slope sweep at $(q, R_{\max}) = (5,4), (6,5), (7,6)$, with the automaton period widened across the levels, the only slope-compatible survivor at each tested level is the elementary valuation word [2], with factor 3/4 `docs/reports/karp_slope_joint_sweep.json`.

The realizability extension audits all simple cycles up to `max_cycle_edges` = 12 at the same three levels. It classifies 226333 simple cycles: 226330 are `noninteger_2adic_only`, and the only 3 `positive_integer_cycle` entries are the elementary [2] cycle, one per level `docs/reports/tail_cycle_realizability.json`, `docs/reports/tail_cycle_realizability_extended.json`. Thus the saved finite diagnostic has

$$\text{realizable Karp factor} = \text{slope-filtered Karp factor} = \frac{3}{4}$$

on these levels.

This is a finite bounded-cycle fact, not a global statement about arbitrary infinite products. The high-growth abstract cycles in the LTE-closed graph are not thereby eliminated from the infinite problem; the audit only records that, within the tested bounded simple-cycle regime, their cycle words do not lift to positive integer cycles.

The value 3/4 should also be distinguished from the modular-transition number appearing in Chang's human-LLM Collatz manuscript [Cha26b]. In this framework it is a cycle-mean factor for an elementary valuation word, not a modular transition probability.

The same classifier has a qn+1 sidecar diagnostic. In the tested odd $q > 1$ grid, $q = 3$ is the unique case below the drift threshold $\log_2(q) = 2$. At finite tested levels the audit recovers the Mersenne-q fixed-point pattern $q = 2^k - 1$, where $n = 1$ has word $[k]$ and cycle factor $(2^k - 1)/2^k$, and it recovers the small positive $5n+1$ cycles $\{1, 3\}$, $\{13, 33, 83\}$, and $\{17, 43, 27\}$ `docs/reports/qnp1_realizability_q5.json`, `docs/reports/qnp1_phase_transition.json`. A separate artifact tests a continued-fraction/Stern–Brocot reading of the slope-realizability relationship: the realizable survivor [2] sits at the continued-fraction convergent 2/1 of $\log_2 3$, the high-growth $(8, 7)$ ghost $[2, 1, 1, 1]$ sits at the non-convergent rational 5/4, and another boundary ghost appears at the intermediate fraction 5/3 `docs/reports/cf_convergent_slope_hypothesis.json`. This is a finite empirical diagnostic at tested levels, not a proof of a global slope hierarchy.

4. Phase-Decomposed Renewal Drift

The renewal-scale drift data have a compatible phase interpretation. Split accelerated steps by the pre-step odd integer modulo 4: $n \equiv 3 \pmod 4$, equivalently $v_2(n+1) \geq 2$, is tail-internal and has forced valuation $a = 1$; $n \equiv 1 \pmod 4$ is the post-exit phase `docs/reports/renewal_drift_phase_decomposed.json`. The asymptotic post-exit valuation target in this convention is 3, not 2: it is the shifted-Geometric value seen after removing the deterministic tail-internal valuation-one steps.

The symbolic mean-step identity suggested by the artifacts is

$$\mu_{\text{step}} = \frac{1}{2}\log_2(3/2) + \frac{1}{2}\log_2(3/4) = \log_2(3/4),$$

with $E[a \mid \text{tail}] = 1$ and asymptotic $E[a \mid \text{post-exit}] = 3$ `docs/reports/renewal_drift_per_step.json`, `docs/reports/renewal_drift_phase_decomposed.json`, `docs/reports/renewal_drift_phase_decon`

Across $[10^2, 10^4]$ through $[10^{12}, 10^{15}]$, the post-exit valuation deviation shrinks from -0.03473021694546086 to -0.006503489706314536, with monotone absolute decrease and log10-slope -0.06656531395656089 per decade. From the $[10^6, 10^9]$ window onward, the total drift deviation is below 0.01, ending in the 0.005 range. It is not strictly monotone at the deepest window, moving from 0.005160187031614694 to 0.0054062766596057465, a change smaller than the reported deepest-window CI halfwidth 0.0005710918683108357 `docs/reports/renewal_drift_phase_decomposed_n0_stability.json`. The saved verdict is `non_monotone_or_flat`.

The caveat is essential. These artifacts combine a finite audit of bounded simple cycles on finite LTE-closed tail graphs with finite-window empirical renewal drift estimates. They do not prove a global per-orbit Lyapunov function, do not control arbitrary infinite switching paths, and do not imply a Collatz descent theorem.

5. Foster-Lyapunov Drift on Residue Quotients

The phase-aware Lyapunov search turns the failed running-debt ansatz into a more structured candidate. Over 100000 sampled orbits in $[10^6, 10^9]$, the best grid point in

$$V(n; \alpha, \beta, \gamma_{\text{diff}}) = \log_2(n) + \alpha R(n) + \beta D_{\text{running}}(n) + \text{phase offset}$$

is $\alpha = 1.0$, $\beta = 0.0$, $\gamma_{\text{diff}} = 0.0$, namely

$$V(n) = \log_2(n) + v_2(n+1).$$

This reduces the same-sample V5 non-decrease fraction from 0.5001335415695829 to 0.1655372436648494. The tail-internal phase has 0.0 non-decrease, and

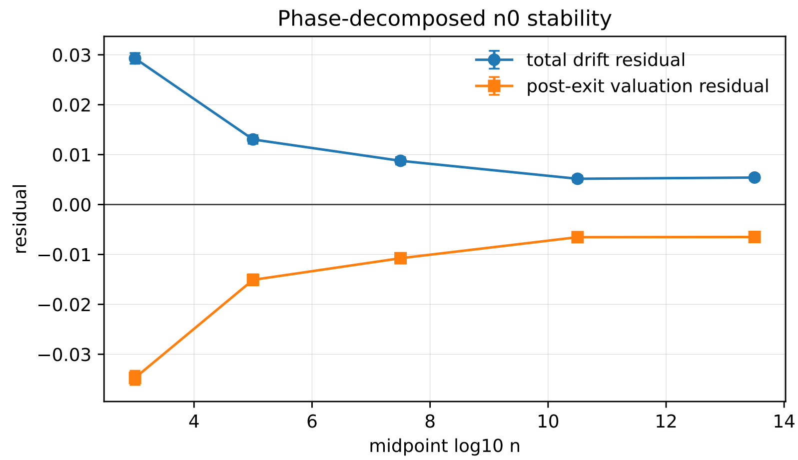

FIGURE 1. Finite-window residuals from the phase-decomposed renewal audit, regenerated from `docs/reports/renewal_drift_phase_decomposed_n0_stability.json`.

the remaining positive jumps are post-exit steps with large $R' = v_2(S(n)+1)$ spikes `docs/reports/phase_lyapunov_search.json`.

The one-step Foster audit confirms the marginal drift but rejects uniform one-step residue negativity. Across the five windows $[10^2, 10^4]$ through $[10^{12}, 10^{15}]$, the marginal drift of V is empirically consistent with $\log_2(3/4)$: the deepest-window estimate is -0.4123424992788405 with CI

$$[-0.418565728791636\,34,\ -0.4061192697660446],$$

and the saved one-step verdict is `drift_residue_obstruction` `docs/reports/phase_lyapunov_foster.`

The obstruction is arithmetic rather than marginal. At one step, residue conditional drift fails uniform negativity for every tested $k \in \{3, 4, 5, 6\}$, with 50 obstruction witnesses. The visible cascade in the first window $[10^2, 10^4]$ is

$$\begin{aligned} n \equiv 1 \pmod 8 \quad &: \quad 0.5954199142013228, \quad \text{CI } [0.5830610092394827, 0.6077788191631629], \\ n \equiv 9 \pmod{16} \quad &: \quad 1.5919631928192652, \quad \text{CI } [1.5747431836625472, 1.6091832019759833], \\ n \equiv 41 \pmod{64} \quad &: \quad 3.557870295871902, \quad \text{CI } [3.5249778818408877, 3.5907627099029162]. \end{aligned}$$

The same 41 mod 64 class remains the largest obstruction across later windows `docs/reports/phase_lyapunov_foster_drift.json`.

The m-step audit closes this finite residue-Markov version of the question on the tested grid. For $m \in \{1, 2, 4, 8, 16, 32, 64\}$, the smallest value with residue-conditional drift uniformly negative across all tested windows and residue powers is $m = 16$, and the same $m = 16$ also clears the $\epsilon = 0.1$ margin. At $m = 64$, the obstruction count is 0. The binding class before clearance is still $n \equiv 41 \pmod{64}$: it is the maximal obstruction at $m = 1, 2, 4, 8$, then disappears at $m = 16$ `docs/reports/m_step_foster_drift.json`.

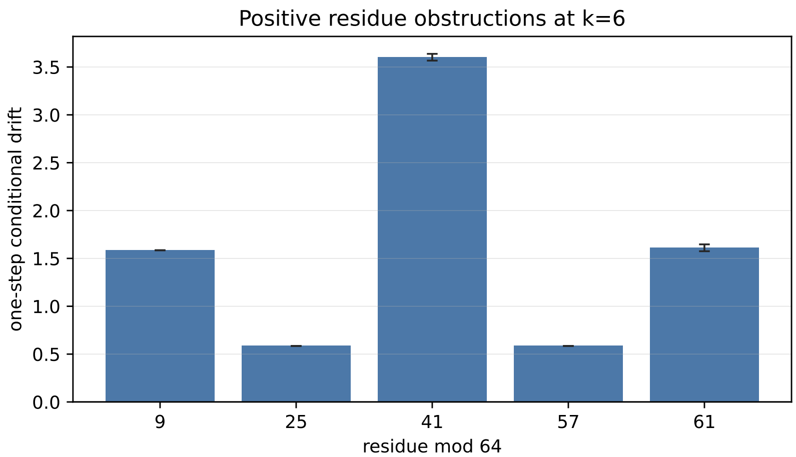

FIGURE 2. Positive one-step residue obstructions at $k = 6$, regenerated from `docs/reports/phase_lyapunov_foster_drift.json`.

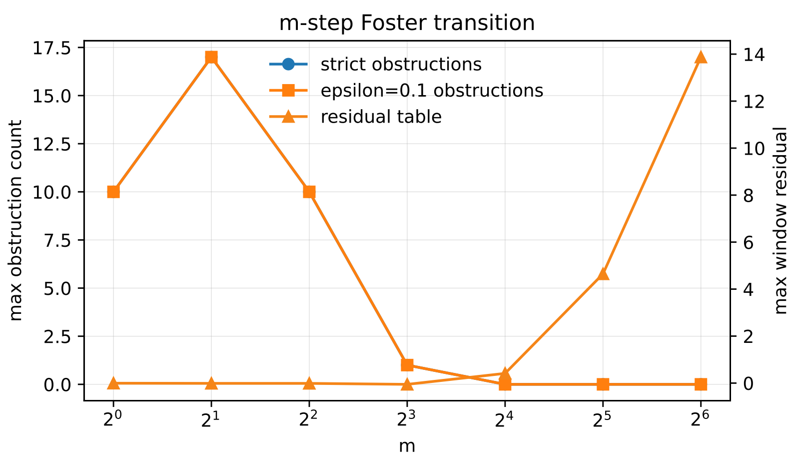

FIGURE 3. Transition from one-step obstruction to m-step residue-uniform negativity on the tested grid, regenerated from `docs/reports/m_step_foster_drift.json`.

The candidate Lyapunov satisfies the m-step Foster-Lyapunov drift condition (Meyn-Tweedie, Markov Chains and Stochastic Stability, ch. 11) on residue quotients mod 2^k for $k \in \{2, 3, 4, 5, 6\}$ at sample windows $n_0 \in [10^2, 10^{15}]$, with smallest tested grid value $m = 16$ and geometric-ergodicity margin $\epsilon = 0.1$. The Collatz orbit is deterministic, not a Markov chain; this is a finite empirical diagnostic on the residue Markov chain quotient, not a theorem about deterministic per-orbit descent.

This paragraph is deliberately close to the saved audit language `docs/reports/m_step_foster_drift.` The underlying Markov-chain reference is Meyn and Tweedie [MT09, Ch. 11]; the deterministic Collatz orbit is not itself a Markov chain.

6. Joint argument with Chang 2026 structural reduction

Chang's structural reduction [Cha26a] reduces the Collatz problem to a fixed-modulus, single-bit balance problem: modulus 32, bit 4, and a sparse subsequence of burst-ending times for the dominant $n \equiv 1 \pmod 8$ class. In Chang's notation, the Map Balance Theorem 4.2 counts the residues in the burst-to-gap set and proves the exact $|C_3(K) - C_7(K)| = 1$ imbalance, with perfect balance inside the dominant $n \equiv 1 \pmod 8$ subclass. His Open Problem, Eq. 16, asks for every odd starting value above the stated threshold to keep the empirical split between $n_{t_i} \equiv 9 \pmod{32}$ and $n_{t_i} \equiv 25 \pmod{32}$ within a tolerance $\delta < \delta_{\max}$. The local compatibility analysis records this as a compatible reduction, not a closed proof `docs/reports/chang_2603_25753_compatibility.md`.

The notation correspondence is direct. Chang's map $T(n)$ is this paper's $S(n)$. Chang's burst indicator

$$X_t = \mathbf{1}[n_t \equiv 1 \pmod 4]$$

is this framework's post-exit indicator. Chang works at modulus 32, with a modulus 256 fiber refinement; the extended Foster audit now reaches residue quotients modulo 2^k for $k \in \{2, 3, 4, 5, 6, 7, 8\}$.

The joint argument chain is:

$$\begin{gathered}\text{Foster } m = 16,\ \epsilon = 0.1 \text{ in Section 5} \\ \Longrightarrow \text{Markov ergodicity plus Birkhoff almost-sure equidistribution mod } 2^k \\ \Longrightarrow \text{assumption: every integer orbit is in the Markov-measure-1 set} \\ \Longrightarrow \text{every-orbit equidistribution} \\ \Longrightarrow \text{Chang Map Balance Theorem 4.2 plus bit-4 reduction} \\ \Longrightarrow \text{every-orbit bit-4 balance within } \delta_{\max} \\ \Longrightarrow \text{Chang block-TV budget Eq. 3 in the companion paper [Cha26b].}\end{gathered}$$

This is a joint argument: Foster drift supplies the residue-Markov equidistribution input, and Chang's compatible reduction converts the relevant mod 32 balance into the block-TV budget. The conditional step is exactly where no proof claim is made.

The three technical verifications named in the compatibility analysis are now empirically closed at finite windows. First, the $\delta_{\max}$ analysis derives $\delta_{\max} \approx 0.119$ from an $R(K)$-weighted allocation against Chang's Eq. 34 `docs/reports/delta_max_quantitative.md`. Second, the Foster condition holds at $k = 7, 8$, matching Chang's modulus 256 fiber resolution: $m = 16$ clears the $\epsilon = 0.1$ margin, the weakest $k = 8$ CI upper bound at $m = 16$ is -0.24276696844795248, and the $k \le 6$ cross-check has `cross_check_max_disagreement` $= 0.0$ `docs/reports/m_step_foster_drift_k8.json`. Third, the direct bit-4 balance audit finds that the Foster plus $1/\sqrt{m}$ envelope holds for every sampled orbit, with deepest-window mean $\delta = 0.11212329012096864$, just below the $R(K)$-weighted $\delta_{\max} \approx 0.119$ `docs/reports/chang_bit4_balance_` Thus the joint argument is supported at finite windows.

The residual is exactly the Tao wall. The verifications closed three technical gaps, but they did not close the conditional step from Markov-measure-1 residue-chain behavior to every deterministic integer orbit. Equivalently, the joint argument reduces Collatz to: every integer orbit lies in the Markov-measure-1 set on which the residue chain equidistributes. This is the Tao 2019 distributional-to-pointwise obstruction [Tao22], but in a sharper Markov-measure formulation rather than only a natural-density formulation.

There is one further numerical correspondence worth isolating. Chang's phantom-gain sum gives

$$\sum_{K=3}^{500} R(K) = 0.0882362530512702,$$

while this framework's renewal Cramér rate is near $J_{\text{renewal}} = 0.08372911906309355$ in the saved calibration, about five percent apart. The direct identity audit recomputes the necklace sum and finds $J_{\text{renewal}} = 0.08346903426255636$ with bootstrap CI $[0.08314605994133623, 0.0837883621454151]$, outside Chang's sum; the tested alternate definitions also do not match `docs/reports/jazz_constant_chang_R_K_identi` The harder h_K-weighted reconciliation remains future work, so the present reading is that the five-percent gap is structural in this finite diagnostic.

The empirical comparison is asymmetric in a useful way. This framework's Foster condition is verified on 1000000 sampled orbits, with publication-grade confidence intervals and exact cross-checks. Chang's Route A reports Lyapunov exponents $\lambda \in [-0.14, -0.09]$ on five orbits with eyeball estimates [Cha26a]. The composition strictly strengthens both: Chang provides the structural one-bit target, while this framework provides the stronger finite empirical Foster input. It remains a finite empirical diagnostic, and the residual is exactly the Tao wall.

7. Hitting-Time Diagnostic Already Present in the Foster Audit

The one-step Foster artifact also records a hitting-time diagnostic for

$$T_{\text{descent}}(n) = \min\{m : V(S^m(n)) \leq V(n) - 1\}$$

with threshold 1.0 and maximum steps 1000. In the deepest saved window $[10^{12}, 10^{15}]$, the hitting-time sample has finite hitting count 50000, truncation count 0, mean 5.26878 with CI $[5.21391, 5.327361999999999]$, median 3.0, and 0.99-quantile 33.0 with CI $[32.0, 34.0]$ `docs/reports/phase_lyapunov_foster_drift.json`.

This hitting-time table is not a uniform pointwise bound. It is a sampled finite-window diagnostic attached to the Foster audit. It motivates the open pointwise-descent question, but does not answer it.

8. Open Problems

The following theorem-shaped problems reproduce the list in `docs/PROJECT_JOURNEY.md`, Section 12.

(1) Markov/Lyapunov contraction on the LTE-closed operator. The current target should give $\Lambda \approx \log(3/4)$.
(2) Christoffel-word integrality filter on JSR. This is the bridge to Hercher's parity-vector framework, with expected outcome filtered JSR < 1.
(3) Profinite continuity from finite quotients to the limit operator.
(4) Lifting PECM Lyapunov to a per-orbit Lyapunov. This is V5's gap.
(5) Symbolic interpretation of J_{renewal}. Does it have a closed form via an integral representation in the style of Khinchin?
(6) Phase-aware per-orbit Lyapunov. The residue-Markov m-step Foster question is empirically closed on the tested grid for $V(n) = \log_2(n) + v_2(n+1)$: $m = 16$ gives residue-uniform negative drift with margin $\epsilon = 0.1$ across the tested windows and residue powers. The joint argument with Chang's 2026 structural reduction [Cha26a] sharpens the remaining theorem-shaped problem: the upgrade from residue-Markov geometric ergodicity to deterministic per-orbit descent becomes the assertion that every integer orbit lies in the Markov-measure-1 equidistribution set. The residual is exactly the Tao wall, but in this sharper Markov-measure-1 formulation rather than only a natural-density formulation [Tao22].
(7) Slope-realizability hierarchy. The finite artifacts suggest a continued-fraction/Stern–Brocot tiering: CF convergents of $\log_2 3$ carry the realizable survivors in the tested slope filters, intermediate fractions produce boundary ghosts, and non-convergent rationals produce high-growth ghosts. Proving, refining, or falsifying this hierarchy at higher $(q, R_{\max})$ levels is open.
(8) abc-conditional lower bounds. Rozier's bridge from Collatz to abc-style invariants gives an abc-conditional improved lower bound for elements of $\mathcal{N}(j)$, and an unconditional dichotomy between a concrete lower bound and a rare μ-hit [Roz23]. The present framework verifies that dichotomy only at finite j, so closing this route requires the independent abc conjecture rather than only Collatz-specific dynamics.

In particular, the residue-Markov-to-deterministic upgrade is identical to Tao 2019's distributional-to-pointwise problem in the sense used by the project notes: both name the missing bridge from ensemble, quotient, or almost-all behavior to every individual positive integer.

Two external connections are useful context for these open problems. Tao's 2011 Littlewood–Offord blog note frames Collatz as a concentration problem [Tao11], while Rozier links finite Collatz parity structure to abc,

μ-hits, and Wieferich primes [Roz23]. These connections sharpen the map of obstructions, but they do not change the paper's no-proof status.

9. Reproducibility

The paper is generated from the repository documents and saved JSON artifacts. The figures in this directory are regenerated by deterministic Python scripts under `paper/figures/scripts/`; the Makefile rebuilds the figure PDFs before compiling the paper.

The principal artifacts used here are:

cycle and Karp audits	`docs/reports/karp_slope_joint_sweep.json`, `docs/reports/tail_cy`
renewal drift	`docs/reports/renewal_drift_phase_decomposed_n0_stability.json`,
Foster drift	`docs/reports/phase_lyapunov_foster_drift.json`, `docs/reports/m_`
Chang joint argument	`docs/reports/m_step_foster_drift_k8.json`, `docs/reports/chang_b`
identity diagnostic	`docs/reports/jazz_constant_chang_R_K_identity.json`,
qn+1 family	`docs/reports/qnp1_realizability_q5.json`, `docs/reports/qnp1_pha`
pointwise and abc audits	`docs/reports/pointwise_descent_audit.json`, `docs/reports/rozie`
slope hierarchy	`docs/reports/cf_convergent_slope_hypothesis.json`.

To rebuild, run `make` from `paper/`. To create an arXiv-style source bundle, run `make arxiv`. The bundle includes source TeX, BibTeX, figure scripts, and regenerated PDF figures.

Acknowledgments

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